

CS173: Discrete Structures

Notes

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Divisibility

Definition. Given $x, y \in \mathbb{Z}$, we define

$$x \mid y \iff (\exists z \in \mathbb{Z})(x \cdot z = y).$$

We read $x \mid y$ as “ x divides y ”. For example, $3 \mid 6 \iff (\exists z \in \mathbb{Z})(3 \cdot z = 6) \iff 3 \cdot 2 = 6$.

Definition. Given $x, y \in \mathbb{Z}$, we define

$$x - y := x + (-y).$$

Definition. We define $|\cdot| : \mathbb{Z} \rightarrow \mathbb{N}$ given by, for every $x \in \mathbb{Z}$:

$$|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Examples.

- $2 \mid 0 \iff (\exists z \in \mathbb{Z})(2 \cdot z = 0)$, witnessed by $z = 0$ since $2 \cdot 0 = 0$.
- $0 \nmid 2 \iff \neg(\exists z \in \mathbb{Z})(0 \cdot z = 2) \iff (\forall z \in \mathbb{Z})(0 \cdot z \neq 2)$, which holds since $0 \cdot z = 0 \neq 2$ for all z .

Divisibility is a Partial Order on $\mathbb{N}$

Theorem. Divisibility on $\mathbb{N}$ is a partial order:

- **Reflexivity:** $(\forall x \in \mathbb{Z})(x \mid x)$.
- **Antisymmetry (up to sign):** $(\forall x, y \in \mathbb{Z})((x \mid y \wedge y \mid x) \Rightarrow |x| = |y|)$.
- **Transitivity:** $(\forall x, y, z \in \mathbb{Z})((x \mid y \wedge y \mid z) \Rightarrow x \mid z)$.

Proof. We prove transitivity. Let $x, y, z \in \mathbb{Z}$ and assume $x \mid y$ and $y \mid z$. So there exist $k_1 \in \mathbb{Z}$ and $k_2 \in \mathbb{Z}$ such that $x \cdot k_1 = y$ and $y \cdot k_2 = z$. Then $(x \cdot k_1) \cdot k_2 = z$, therefore $x \cdot (k_1 \cdot k_2) = z$, where $k_1 \cdot k_2 \in \mathbb{Z}$. Hence $x \mid z$. $\square$

Theorem.

- $(\forall x \in \mathbb{Z})(x \mid 0)$.
- $(\forall x \in \mathbb{Z})(1 \mid x)$.
- $(\forall x \in \mathbb{Z})(x \mid x)$.
- $(\forall x, y \in \mathbb{Z})(x \mid y \Rightarrow |x| \leq |y|)$.

Primality

Definition. We say $x \in \mathbb{N}$ is **prime** if

$$(x \geq 2) \wedge (\forall p \in \mathbb{N})(p \mid x \Rightarrow p \in \{1, x\}).$$

Example.

- 1 is not prime since $1 < 2$.
- 4 is not prime since $2 \mid 4$ but $2 \notin \{1, 4\}$.

Theorem. $\mathbb{N} \subseteq \mathbb{Z}$ (false, but spiritually true).

Properties of $(\mathbb{N}, +, \cdot)$

- 0 is the $+$ identity.
- $+$ is associative.
- $+$ is commutative.
- 1 is the $\cdot$ identity.
- $\cdot$ is associative.
- $\cdot$ is commutative.
- $\cdot$ distributes over $+$.
- 0 is the annihilator of $\cdot$.

Strong Induction

Theorem (Strong Induction). Let φ be a predicate on $\mathbb{N}$. If

$$\varphi(0) \wedge (\forall k \in \mathbb{N}) \left((\forall \ell \in \mathbb{N})(\ell < k \Rightarrow \varphi(\ell)) \Rightarrow \varphi(k) \right)$$

then $(\forall n \in \mathbb{N})(\varphi(n))$.

The bracketed part $(\forall \ell \in \mathbb{N})(\ell < k \Rightarrow \varphi(\ell))$ is the **inductive hypothesis** — we assume φ holds for *all* values strictly less than k , not just the immediate predecessor.

Fundamental Lemma of Arithmetic

Theorem (Fundamental Lemma of Arithmetic).

$$(\forall n \in \mathbb{N}) \left(n \geq 2 \Rightarrow (\exists p \in \mathbb{N})(p \text{ is prime} \wedge p \mid n) \right)$$

Proof. Base case: Observe $2 \mid 2$ because $2 \cdot 1 = 2$ and $1 \in \mathbb{Z}$. Exercise: NTS 2 is prime.

Inductive step: Let $k \in \mathbb{N}$. Assume

$$(\forall \ell \in \mathbb{N}) \left(\ell < k \Rightarrow (\ell \geq 2 \Rightarrow (\exists p \in \mathbb{N})(p \text{ is prime} \wedge p \mid \ell)) \right).$$

Assume $k \geq 2$.

Case 1: Suppose k is prime. Then $k \mid k$, and k is prime, so k itself is the required prime divisor.

Case 2: Suppose k is not prime (i.e., k is composite). Then there exists $p \in \mathbb{N}$ such that $p \mid k$ and $p \notin \{1, k\}$. So $p \neq 1$ and $p \neq k$. Notice $p \neq 0$ because if $p = 0$, then $p \mid k$ would imply $k = 0$, contradicting $k \geq 2$. So $p \geq 2$. Also, since $p \mid k$, we have $|p| \leq |k|$, so $p \leq k$, and since $p \neq k$, we get $p < k$. By the inductive hypothesis, there exists $q \in \mathbb{N}$ such that q is prime and $q \mid p$. We know $q \mid k$ by transitivity of divisibility.

Therefore $(\forall n \in \mathbb{N})(n \geq 2 \Rightarrow \exists z \in \mathbb{N}(z \mid n \wedge z \text{ is prime}))$. $\square$

Fundamental Theorem of Arithmetic

Theorem (Fundamental Theorem of Arithmetic). Every $n \in \mathbb{N}$ with $n \geq 2$ has a unique prime factorization:

$$(\forall n \in \mathbb{N}) \left(n \geq 2 \Rightarrow (\exists! k \in \mathbb{N}^+)(\exists! p_1, p_2, \dots, p_k \in \mathbb{N})(\exists! \alpha_1, \alpha_2, \dots, \alpha_k \in \mathbb{N}^+) \right. \\ \left. \left(n = \prod_{i=1}^k p_i^{\alpha_i} \right) \wedge (p_1 \text{ is prime}) \wedge (p_2 \text{ is prime}) \wedge \dots \wedge (p_k \text{ is prime}) \right)$$

GCD and Coprimality

Definition (GCD). Given $x, y \in \mathbb{Z}$, we say that $d \in \mathbb{N}$ is the **greatest common divisor** of x and y , written $\gcd(x, y) = d$, if:

1. $d \mid x$.
2. $d \mid y$.
3. $(\forall z \in \mathbb{N})(z \mid x \wedge z \mid y \Rightarrow z \mid d)$.

Definition (Coprime). We say $x, y \in \mathbb{Z}$ are **coprime** if $\gcd(x, y) = 1$.
Theorem.

$$(\forall x, y \in \mathbb{Z}) \left(\gcd(x, y) = 1 \iff (\forall p \in \mathbb{N})(p \text{ is prime} \Rightarrow (p \nmid x) \vee (p \nmid y)) \right)$$

Lemma.

1. $(\forall x, y \in \mathbb{Z})(\gcd(x, y) = 0 \iff x = 0 \wedge y = 0)$.

2. $(\forall x, y \in \mathbb{Z})(\gcd(x, y) = x \iff x \mid y)$.
3. $(\forall x, y \in \mathbb{Z})(x \neq 0 \vee y \neq 0 \Rightarrow \gcd(x, y) \geq 1)$.

Examples.

- $\gcd(1, 9) = 1, \quad \gcd(2, 4) = 2, \quad \gcd(4, 10) = 2$.
- $\gcd(0, 9) = 9$ since $9 \mid 9$. $\gcd(4, 20) = 4 = 2^2$.
- $4 = 2^2, \quad 10 = 2 \cdot 5, \quad 20 = 2^2 \cdot 5$.
- $\gcd(0, 1) = 1, \quad (\forall x \in \mathbb{Z})(\gcd(1, x) = 1)$.

LCM

Definition (LCM). The least common multiple of $x, y \in \mathbb{Z}$ is a natural number $d \in \mathbb{N}$ such that:

1. $x \mid d$.
2. $y \mid d$.
3. $(\forall z \in \mathbb{Z})(x \mid z \wedge y \mid z \Rightarrow d \mid z)$.

Euclid's Division Lemma

Theorem (Euclid's Division Lemma). For any $x, y \in \mathbb{Z}$, if $y \neq 0$, then

$$(\exists! q, r \in \mathbb{Z})(x = q \cdot y + r \wedge 0 \leq r < |y|).$$

Lemma.

$$(\forall x, y, n \in \mathbb{Z}) \left((n \mid x \wedge n \mid y) \Rightarrow (\forall a, b \in \mathbb{Z})(n \mid ax + by) \right)$$

Lemma.

$$(\forall a, b, c \in \mathbb{Z})(a = b + c \Rightarrow \gcd(a, b) = \gcd(a, c) = \gcd(b, c))$$

Algorithm (Euclidean Algorithm). For $a, b \in \mathbb{Z}$:

$$\gcd(a, b) := \begin{cases} a & \text{if } b = 0 \\ \gcd(b, r) & \text{if } b \neq 0, \text{ where } r \in \mathbb{Z} \text{ s.t. } (\exists q \in \mathbb{Z})(a = q \cdot b + r) \wedge 0 \leq r < |b| \end{cases}$$

Note: $\gcd(x, y) = \gcd(x, r)$ where $x = q \cdot y + r$ and $0 \leq r < |y|$, so $\gcd(x, y) = \gcd(y, r)$.

Bezout's Theorem

Theorem (Bezout).

$$(\forall x, y \in \mathbb{Z})(\exists a, b \in \mathbb{Z})(\gcd(x, y) = ax + by)$$

Proof. Let $x, y \in \mathbb{Z}$. WLOG suppose $x \neq 0$ and $y \neq 0$. Consider

$$M := \{ax + by \mid a \in \mathbb{Z} \wedge b \in \mathbb{Z} \wedge ax + by > 0\}.$$

Observe $(\forall m \in M)(m \in \mathbb{Z} \wedge m > 0)$, so $M \subseteq \mathbb{N}$. If $x > 0$, then $x = 1 \cdot x + 0 \cdot y \in M$. If $x < 0$, then $-x = (-1) \cdot x + 0 \cdot y \in M$. So $M \neq \emptyset$.

Therefore, there is some $d \in M$ such that $(\forall m \in M)(d \leq m)$. Since $d \in M$, we know $d = ax + by$ for some $a, b \in \mathbb{Z}$.

We verify the three conditions for $\gcd(x, y) = d$:

Condition 1 — $d \mid x$: By Euclid's Division Lemma, there exist $q, r \in \mathbb{Z}$ such that $x = q \cdot d + r$ and $0 \leq r < d$. Then:

$$x = q \cdot d + r \Rightarrow x = q(ax + by) + r \Rightarrow (1 - qa)x + (-qb)y = r.$$

Towards a contradiction, suppose $r \neq 0$, so $0 < r < d$. Since $r > 0$ and $r = (1 - qa)x + (-qb)y$, we know $r \in M$. But $d \leq r$ by minimality of d , contradicting $r < d$. So $r = 0$, hence $x = q \cdot d$, so $d \mid x$.

Condition 2 — $d \mid y$: By the same argument applied to y , using Euclid's Division Lemma to write $y = q' \cdot d + r'$ with $0 \leq r' < d$, we get $r' = (-q'a)x + (1 - q'b)y \in M$ if $r' \neq 0$, contradicting minimality of d . So $r' = 0$, hence $d \mid y$.

Condition 3 — d is the greatest: Let $z \in \mathbb{N}$ and suppose $z \mid x$ and $z \mid y$. Since $d = ax + by$, by the linear combination lemma $z \mid ax + by$, so $z \mid d$.

Therefore $d = \gcd(x, y)$, and $\gcd(x, y) = ax + by$ for some $a, b \in \mathbb{Z}$. □

Prime Divisibility

Theorem.

$$(\forall p \in \mathbb{N})(\forall a, b \in \mathbb{Z}) \left((p \text{ is prime} \wedge p \mid ab) \Rightarrow (p \mid a \vee p \mid b) \right)$$

Note: this is special to primes. In general, $m \mid ab \not\Rightarrow (m \mid a \vee m \mid b)$. For example, $20 \mid 50 \cdot 2$ but $20 \nmid 2$ and $20 \nmid 50$.